Data Science for
Reliable Decision Making

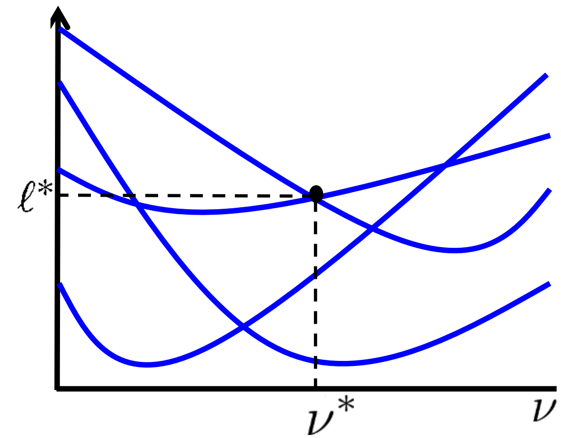
by: *Marco C. Campi* and *Algo Carè*

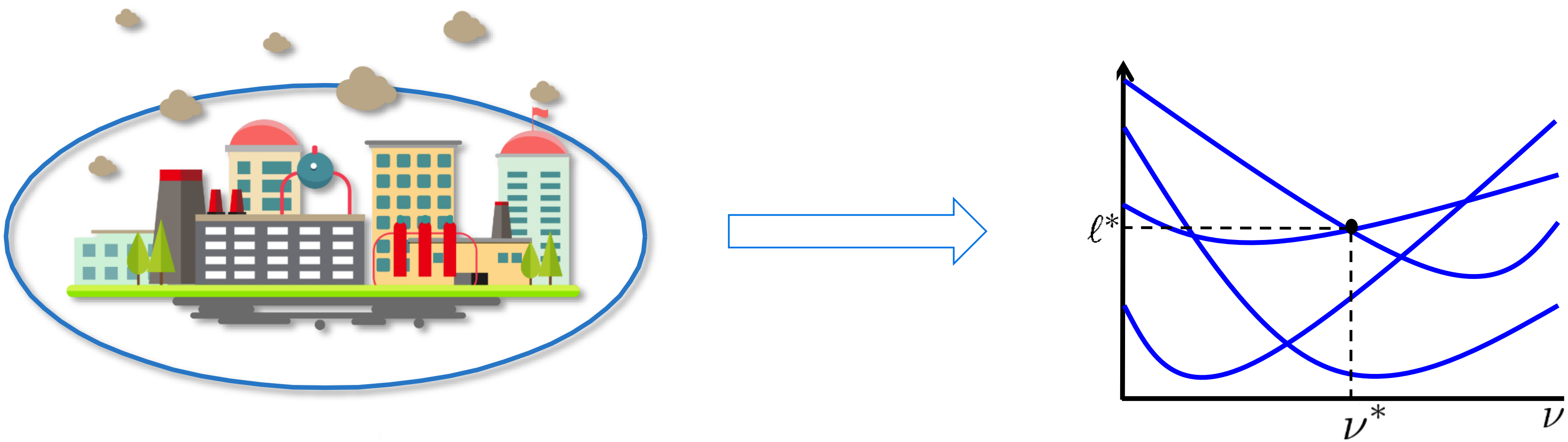
Two key quandaries in data-driven optimization:

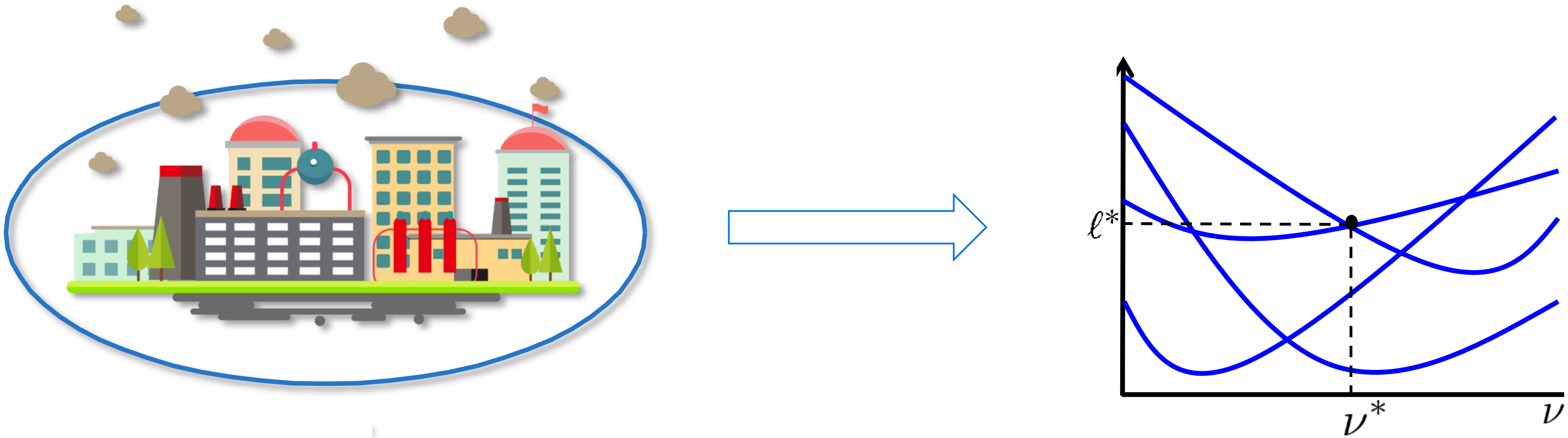
1. indirect or direct?
2. the need for certified data-driven optimization

1. indirect or direct?

real world




ℓ*
ν*
ν


ℓ^*
ν^*
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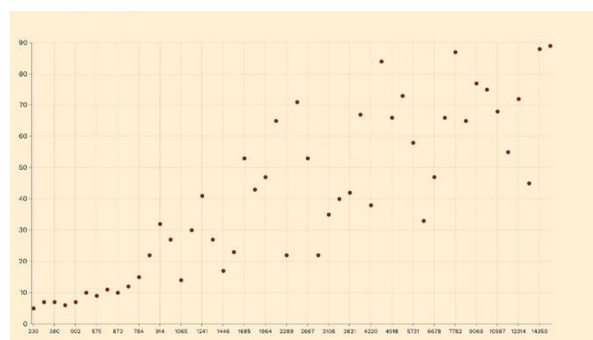
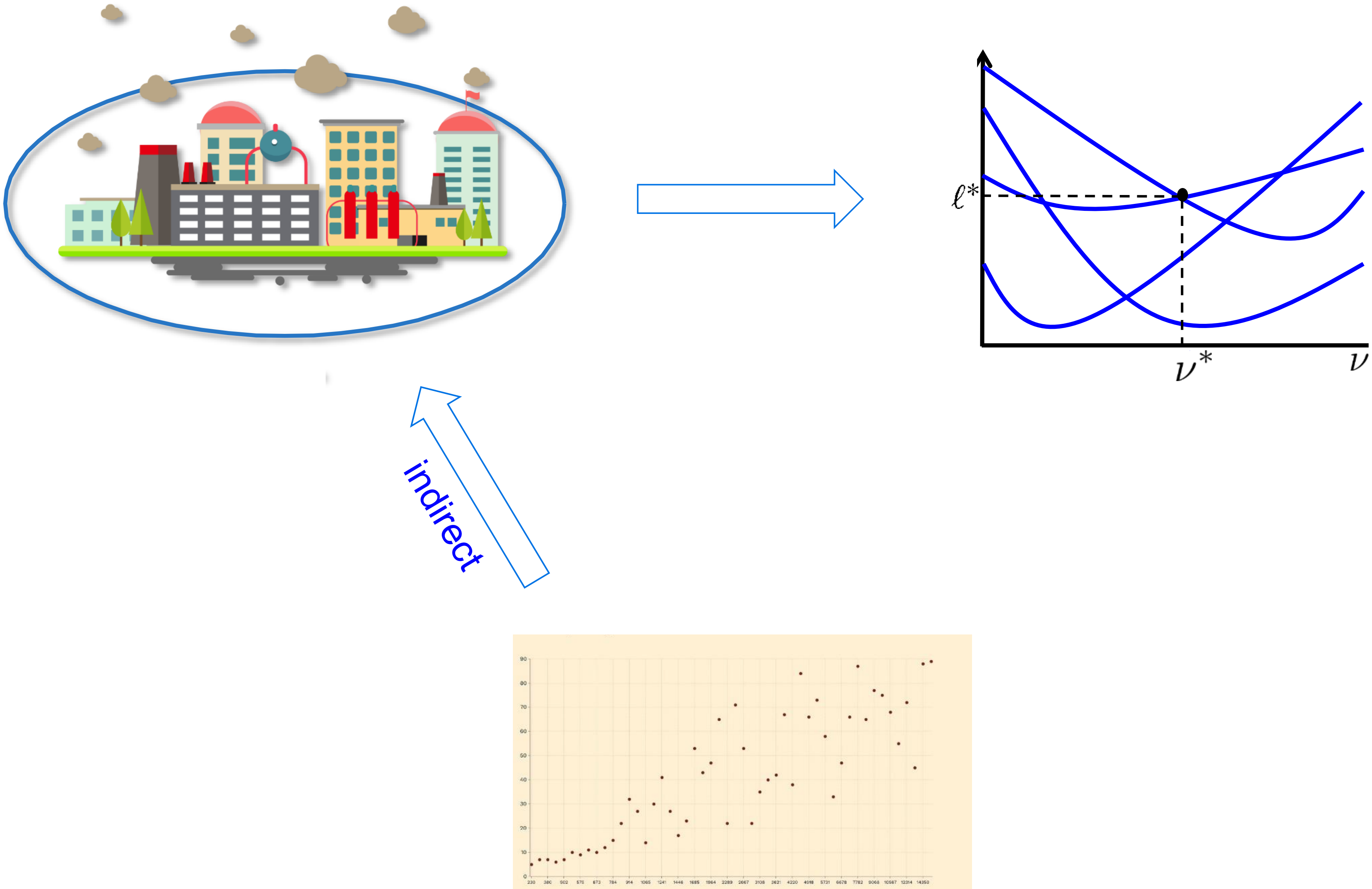
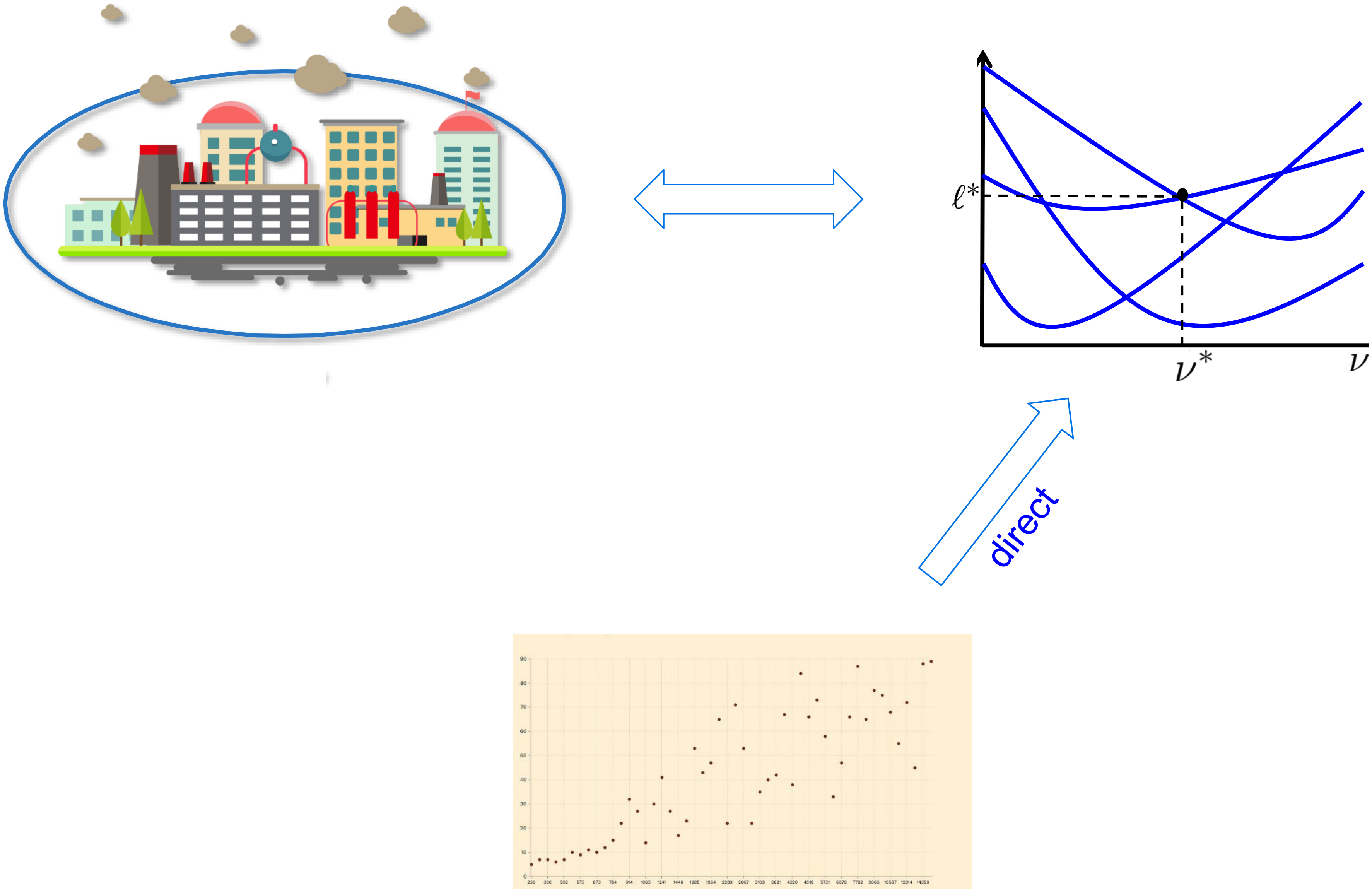



ℓ^*
ν^*
ν
indirect



ℓ^*
ν^*
ν
direct

example: portfolio optimization

q assets

ν^k = percentage of capital invested on asset k

$\nu = \left[\nu^1 \cdots \nu^q\right]^T$

$$\ell(\nu, \delta) = \sum_{k=1}^{q} \nu^k R^k \qquad \delta = [R^1 \ \cdots \ R^q]^T$$

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Black and Scholes → log-normal distribution

q = 100 → 5150 parameters!

example: defibrillation

goal: predict whether the defibrillator shock will be effective

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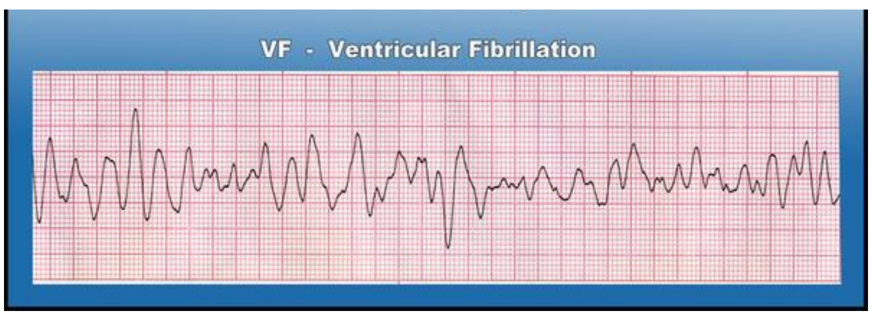
$$\delta_i = (u_i, y_i)$$

$u_i =$ 

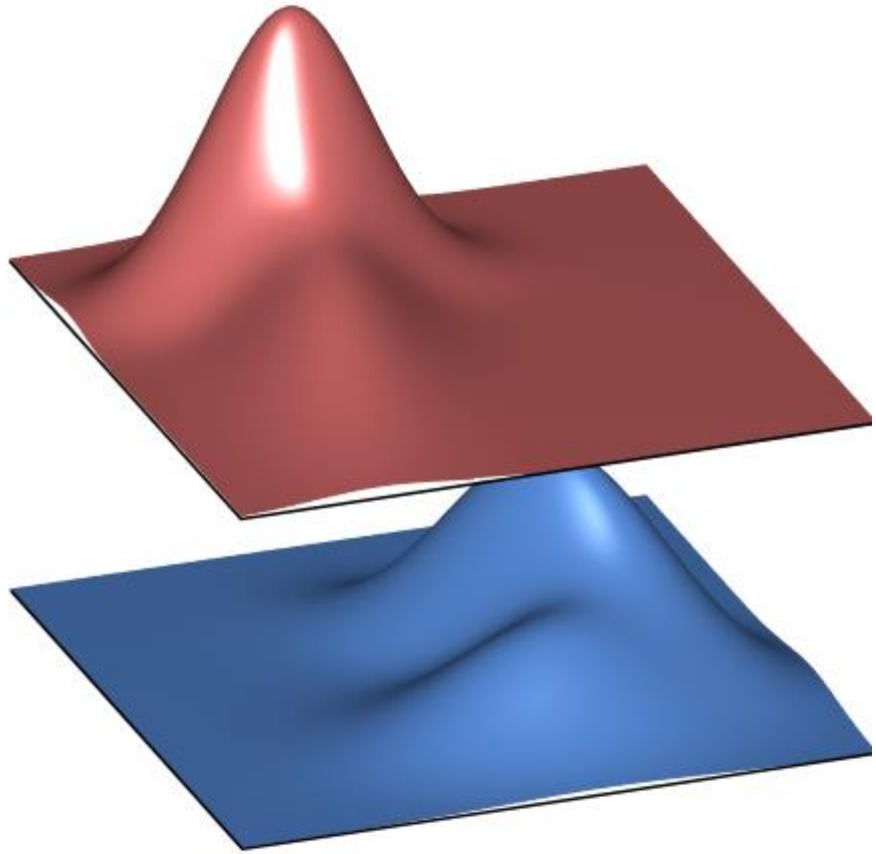
$$y_i = \begin{cases} 0, & \text{uneffective shock} \\ 1, & \text{effective shock} \end{cases}$$

example: defibrillation

ECG = infinite dimentional object → extract finitely many features

- *peak-to-peak (PTP)*
- *maximum amplitude (Amax)*
- *minimum amplitude (Amin)*
- *wave amplitude (WA)*
- *root mean square (RMS)*
- *dominant frequency (DF)*
- *centroid frequency (CF)*
- *edge frequency (EF)*
- *amplitude spectral area (AMSA)*

example: defibrillation



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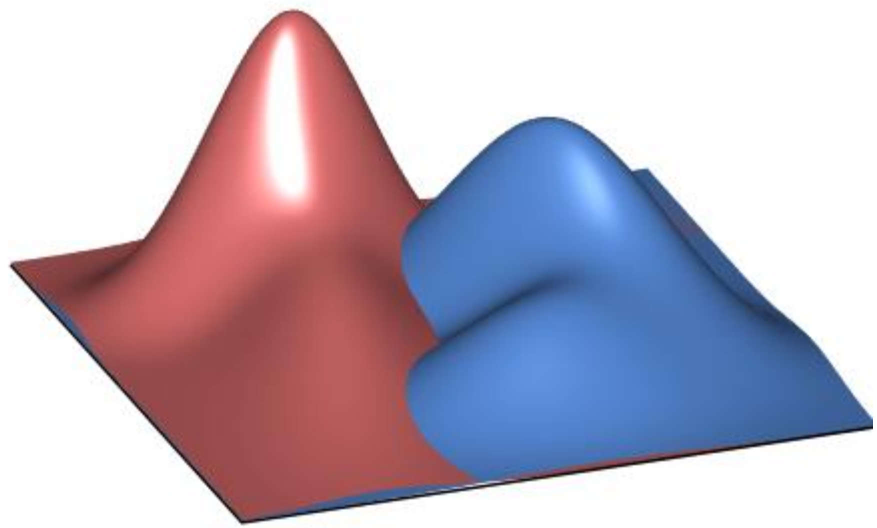
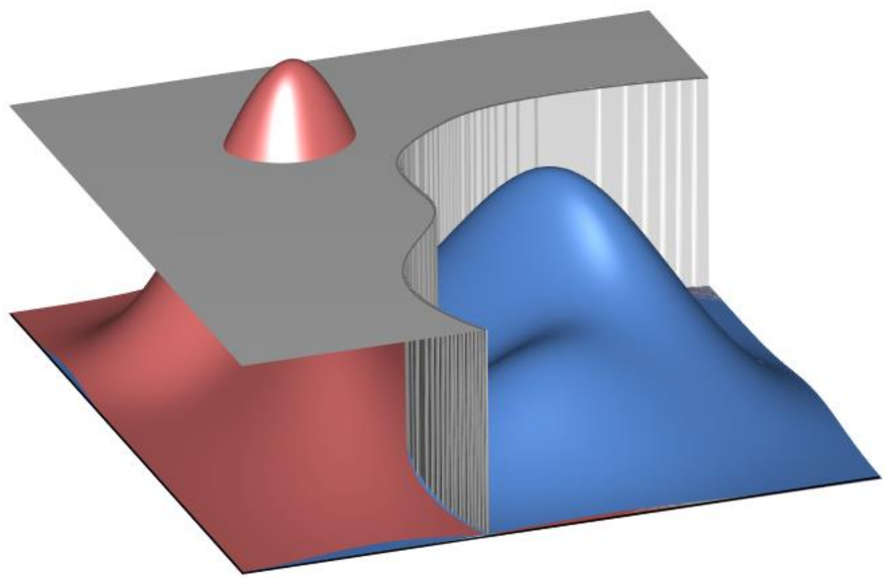


example: defibrillation



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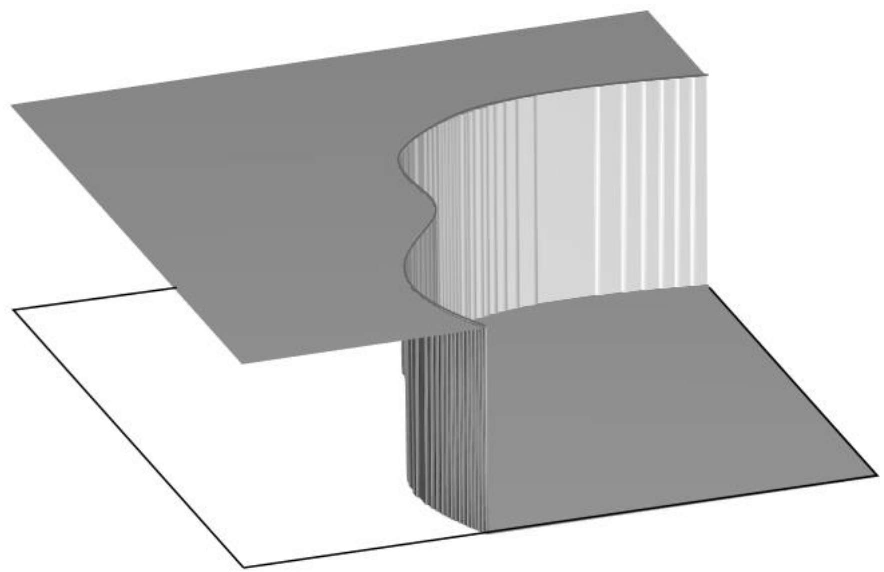

So:

- for simple problems, the indirect path works well
- however, as the complexity of the problem increases, the indirect path falls short
- therefore, in real problems it may be convenient to aim for a direct path: from data to decisions

2. certified data-driven optimization

→ we should aim to develop a science for data-driven optimization

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→ *provide guarantees (certificates of quality)*

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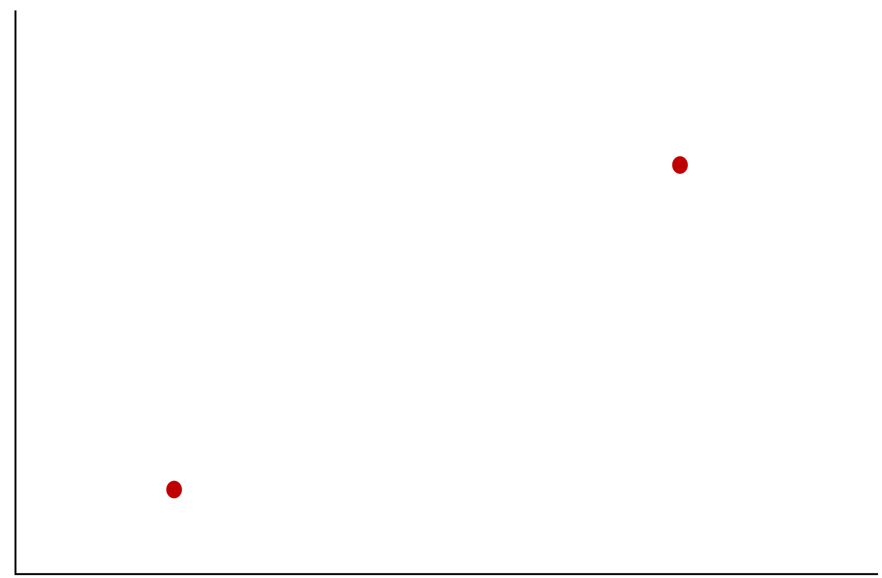

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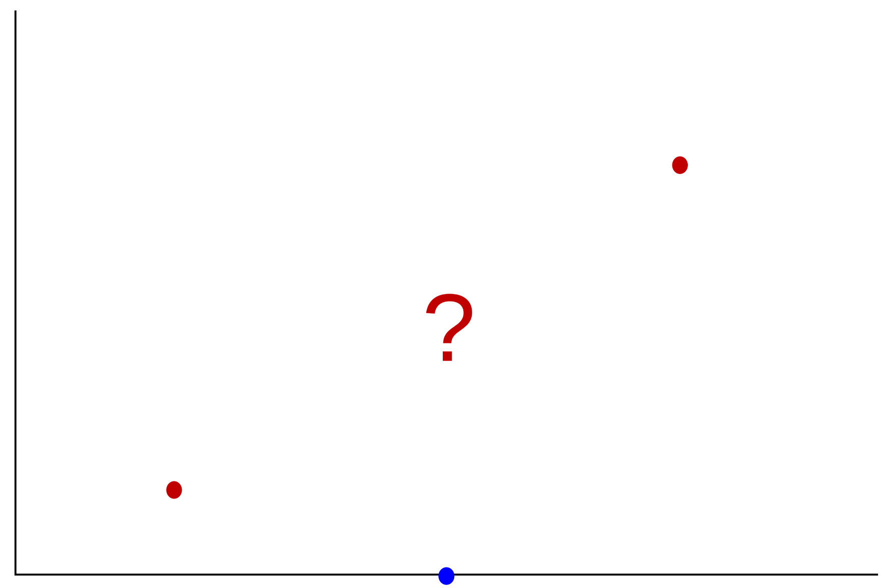

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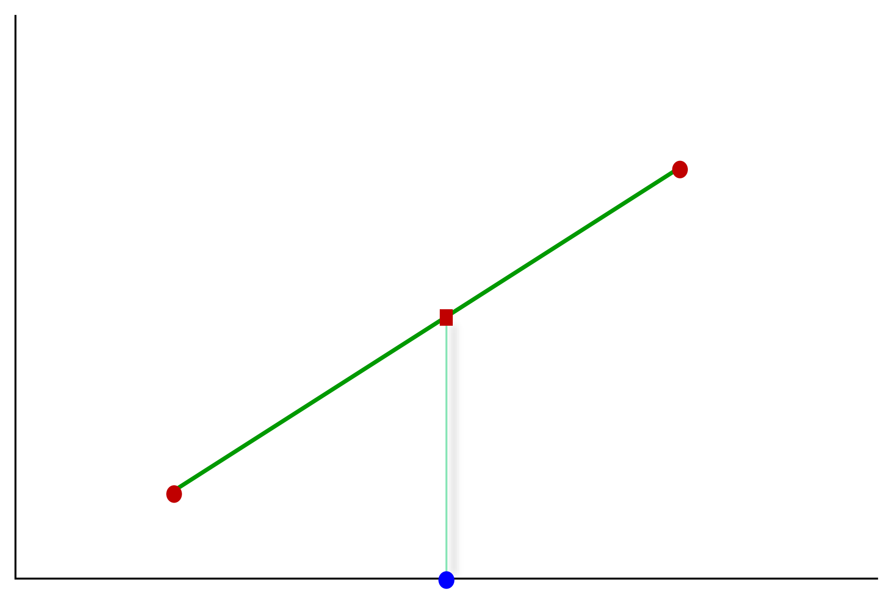

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<u>should we aim higher?</u>

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prior + data → design

~~prior~~ + data → guarantees

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should we aim higher?

prior + data → design → check for features of interest

~~prior~~ + data → guarantees → always remain intact

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is this at all possible?

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→ *provide guarantees (certificates of quality)*

→ *use priors*

<u>should we aim higher?</u>

prior + data → design → check for features of interest

~~prior~~ + data → guarantees → always remain intact

→ *develop trust in the decision*

→ *select among alternatives*

"*The best way to have a good idea is to have lots of ideas*"

Linus Pauling